NX-300

Industrial Compressor System

USER & MAINTENANCE MANUAL | Version 2.0 | September 2024

About this manual

This manual provides operating limits, installation guidance, and routine maintenance instructions for the NexGen NX-300 compressor system. The system is available in Model A and Model B configurations. Confirm the equipment nameplate before connecting power or applying pressure.

READ BEFORE USE

Use the model-specific operating limit in Section 2. Model B's factory-calibration exception is narrow and time-limited; it does not change its normal operating limit.

Document control

Field	Details
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Product family	NX-300 industrial compressor system
Intended audience	Qualified installers, operators, and maintenance technicians
Document status	Fictional sample document for software demonstration and evaluation

Contents

1. Product overview and model identification
2. Technical specifications and operating limits
3. Installation and initial setup
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5. Maintenance and troubleshooting

1. Product Overview

The NX-300 is an industrial compressor intended for controlled air delivery in manufacturing, packaging, and process-automation environments. It includes a digital control panel, pressure monitoring, and automatic safety shutdown.

1.1 Model identification

Feature	Model A	Model B
Priming method	Manual priming valve	Automatic priming
Electrical supply	230 V AC, 50 Hz	400 V AC, 50 Hz
Normal maximum operating pressure	9 bar	12 bar

1.2 Key features

- Digital control panel with live pressure and temperature readings.
- Pressure-relief protection and automatic shutdown on detected fault conditions.
- Model-specific priming procedure to reduce the risk of dry operation.
- Maintenance intervals tracked by operating hours.

2. Technical Specifications

Parameter	Model A	Model B
Maximum normal operating pressure	9 bar	12 bar
Air delivery rate	180 m³/h	300 m³/h
Motor power	15 kW	22 kW
Power supply	230 V AC / 50 Hz	400 V AC / 50 Hz
Inlet connection	DN 40	DN 50
Outlet connection	DN 40	DN 50
Recommended ambient temperature	5°C to 40°C	5°C to 40°C
Minimum clearance	1 m on all sides	1 m on all sides

IMPORTANT — MODEL A LIMIT

Model A must not exceed 9 bar during normal operation. Do not apply Model B's calibration exception to Model A.

3. Installation and Initial Setup

Installation must be performed by qualified personnel. Isolate the equipment from power before connecting or disconnecting pipework. Place the compressor on a stable, level surface in a dry, well-ventilated area.

3.1 Site requirements

- Maintain at least 1 m of clearance around the unit for cooling and service access.
- Do not install where ambient temperature is below 5°C or above 40°C.
- Use pipework and fittings rated for the installed model's maximum pressure.
- Verify the model label and electrical supply before energising the system.

3.2 Model A: manual priming procedure

Step	Action
1	Disconnect and lock out the electrical supply.
2	Confirm inlet and outlet connections are correctly fitted.
3	Open the manual priming valve.
4	Fill the inlet chamber until water appears at the outlet inspection point.
5	Close the manual priming valve and check for leaks.
6	Restore power only after the valve is closed and installation has been inspected.

MODEL A STARTUP CHECK

The manual priming valve must be closed before power is restored. Never run Model A with an unprimed inlet chamber.

3.3 Model B: automatic priming procedure

Step	Action
1	Verify a 400 V AC, 50 Hz supply and correct protective circuit.
2	Connect inlet and outlet pipework; confirm all joints are secure.
3	Check that the inlet path is clear and the unit is correctly grounded.
4	Start using the control panel. The automatic priming cycle fills the inlet chamber.
5	Monitor the panel and stop the unit if a fault indicator or leak appears.

4. Operation, Exceptions, and Safety

4.1 Normal operating limits

Model / condition	Permitted pressure	Rule
Model A — normal operation	Up to 9 bar	Do not exceed the model-specific limit.
Model B — normal operation	Up to 12 bar	12 bar is the normal operating maximum.
Model B — factory calibration only	13 bar for no more than 30 seconds	Authorised factory calibration only.

EXPLICIT EXCEPTION — MODEL B ONLY

Model B may be exposed to 13 bar for a maximum of 30 seconds during factory calibration only. This is not a normal operating condition and does not permit continuous operation at 13 bar. The exception does not apply to Model A.

4.2 Safe operating procedure

- Check the correct model and pressure limit before starting.
- Increase pressure gradually and monitor readings on the control panel.
- Stop immediately if pressure fluctuates unexpectedly, a leak is detected, or an alarm appears.
- Never bypass pressure-relief devices, interlocks, or automatic shutdown features.
- Do not use the calibration exception to justify higher routine operating pressure.

4.3 Shutdown

- Use the control panel to stop the compressor.
- Isolate power before servicing or disconnecting pipework.
- Depressurise the system using the approved site procedure before opening any connection.

The specifications in this fictional manual are for software testing only and are not instructions for real industrial equipment.

5. Maintenance and Troubleshooting

5.1 Routine maintenance schedule

Interval	Inspection / action	Required response
Before each use	Inspect hoses, pipework, fittings, and visible seals.	Do not start if damage or leakage is found.
Every 500 operating hours	Inspect inlet seals and connection gaskets.	Replace damaged seals before restarting.
Every 1,000 operating hours	Inspect control panel, pressure sensor, and safety shutdown.	Refer faults to a qualified technician.
After any overpressure event	Inspect pressure-relief system and record event.	Do not resume until the unit is cleared.

5.2 Troubleshooting guide

Symptom	Possible cause	Action
Pressure does not rise	Restricted inlet; incorrect setup; leak	Stop, isolate power, inspect inlet and connections.
Unexpected pressure spike	Control fault or restricted outlet	Stop immediately and arrange inspection.
Model A fails to prime	Valve not opened or chamber not filled	Repeat Model A priming procedure with power isolated.
Model B priming fault	Inlet restriction or automatic priming fault	Check inlet path; request service if fault persists.
Seal damage or leakage	Wear, incorrect fitting, or excessive pressure	Replace damaged seals before restarting.

5.3 Service information

Record the unit model, serial number, operating hours, fault codes, and any pressure excursion when requesting service. The motor manufacturer is not identified in this manual; consult the equipment nameplate or supplier records for component-specific details.

DOCUMENT LIMITATION

This manual does not identify a motor manufacturer. Do not infer a manufacturer from the NX-300 product name or other specifications.

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